

Heterogeneous Domain Adaptation for IoT Intrusion Detection: A Geometric Graph Alignment Approach

Jiashu Wu^{1b}, Graduate Student Member, IEEE, Hao Dai^{1b}, Graduate Student Member, IEEE, Yang Wang^{1b}, Member, IEEE, Kejiang Ye^{1b}, Member, IEEE, and Chengzhong Xu^{1b}, Fellow, IEEE

Abstract—Data scarcity hinders the usability of data-dependent algorithms when tackling IoT intrusion detection (IID). To address this, we utilize the data-rich network intrusion detection (NID) domain to facilitate more accurate intrusion detection for IID domains. In this article, a geometric graph alignment (GGA) approach is leveraged to mask the geometric heterogeneities between domains for better intrusion knowledge transfer. Specifically, each intrusion domain is formulated as a graph where vertices and edges represent intrusion categories and category-wise inter-relationships, respectively. The overall shape is preserved via a confused discriminator incapable to identify adjacency matrices between different intrusion domain graphs. A rotation avoidance mechanism and a center point matching mechanism are used to avoid graph misalignment due to rotation and symmetry, respectively. Besides, category-wise semantic knowledge is transferred to act as vertex-level alignment. To exploit the target data, a pseudo-label (PL) election mechanism that jointly considers network prediction, geometric property, and neighborhood information is used to produce fine-grained PL assignment. Upon aligning the intrusion graphs geometrically from different granularities, the transferred intrusion knowledge can boost IID performance. Comprehensive experiments on several intrusion data sets demonstrate state-of-the-art performance of the GGA approach and validate the usefulness of GGA-constituting components.

Index Terms—Domain adaptation (DA), geometric graph alignment (GGA), Internet of Things (IoT), intrusion detection, pseudo-label election.

I. INTRODUCTION

INTERNET of Things (IoT) devices become indispensable for various real-world applications and innovatively transform several fields such as healthcare [1], [2], etc. However,

the limited power and computational capability of IoT devices hinder the applicability of powerful security mechanisms, together with relatively infrequent maintenance, making IoT vulnerable to malicious intrusions. Therefore, to keep the IoT infrastructure safe, an effective IoT intrusion detection (IID) system is vital. To advance the intrusion detection techniques, several research directions become popular. Dietz et al. [3] proposed to automatically scan IoT devices for predefined vulnerability patterns, and isolated suspicious devices to block the botnet spreading. McDermott et al. [4] tackled the problem via deep recurrent neural network (RNN) and achieved satisfying detection performance. However, these efforts required either a thorough intrusion pattern repository, or abundant labeled training data, which is expensive to collect and time-consuming to annotate, and is, especially, difficult for IoT due to factors such as data privacy concerns, the frequent emergence of new IoT things, etc. Therefore, the data scarcity of IoT hinders the usability of these rules or the data-dependent methods.

Considering that the Internet intrusion data is richer than IoT domains, and they share several similar intrusion categories, several domain adaptation-based (DA) methods were proposed to treat the network intrusion (NI) as the source domain and transfer rich intrusion knowledge to facilitate the data-scarce target IoT intrusion (II) domains. By achieving domain-invariant alignment, the transferred intrusion knowledge can facilitate more accurate IID. For instance, Ning et al. [5] presented a Laddernet-based DA solution to improve the intrusion classification accuracy and secure the industrial IoT infrastructures. Hu et al. [6] studied a deep subdomain adaptation network with an attention mechanism and focused on distribution alignment between domains via local maximum mean discrepancy. Methods such as [7] and [8] proposed to achieve intrusion domain alignment by aligning the graph learning results. Efforts such as [9], [10], and [11] attempted to explore unlabeled target domain via directly predicted, threshold selected, or softly assigned pseudo-labels (PLs), respectively, to facilitate better intrusion knowledge transfer.

Despite their success, they left some deficiencies that need to be addressed. Previous DA-based ID methods didn't tackle the problem from a geometric graph perspective and failed to explore unlabeled target domain via geometric and neighborhood-aware PLs. The ignorance of intrinsic geometric properties in domain graphs and the under-explored target domain can result in coarse-grained alignment. Although some graph-based DA methods were proposed, they did not attempt

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Jiashu Wu and Hao Dai are with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: wujiashu21@mails.ucas.ac.cn; daihao19@mails.ucas.ac.cn).

Yang Wang and Kejiang Ye are with the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: yang.wang1@siat.ac.cn; kj.ye@siat.ac.cn).

Chengzhong Xu is with the State Key Laboratory of IoT for Smart City, Faculty of Science and Technology, University of Macau, Macau, China (e-mail: czxu@um.edu.mo).

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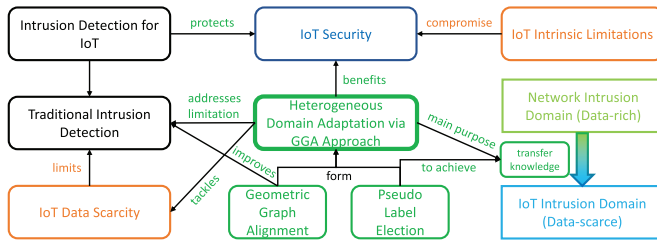


Fig. 1. General motivation of the GGA approach.

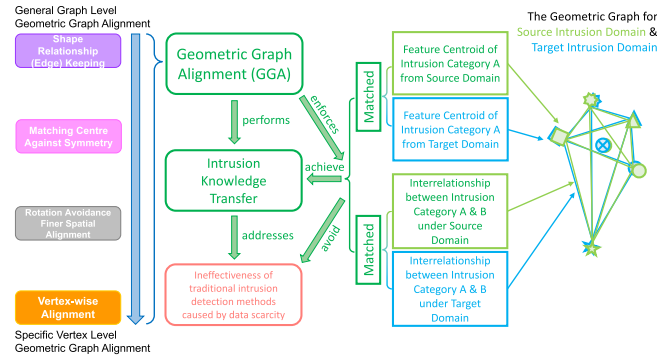


Fig. 2. Overview of the GGA mechanism.

graph alignment from a pure geometric perspective, leaving the geometric properties under-explored. Despite some methods can partially convey the geometric properties through graph embedding, however, the embedding learning is highly data-dependent which is challenging for IoT scenarios. Besides, although there were some traditional PL-based DA methods, their isolated PL assignment strategy failed to leverage the geometric and neighborhood information between labels, which may produce error-prone PLs and mislead the intrusion knowledge transfer.

To address these limitations and achieve more fine-grained intrusion knowledge transfer, following the motivation illustrated in Figs. 1 and 2, we propose a geometric graph alignment (GGA) approach that works under the semi-supervised heterogeneous DA (HDA) setting, i.e., the target is scarcely-labeled and significant source-target heterogeneities exist, such as having diverse feature spaces, following different distributions, etc. To positively exploit the unlabeled target domain, we utilize a PL election (PLE) mechanism. To prevent the error-prone PL from misleading the model, the geometric property is considered to eliminate confident but incorrect PLs based on their geometric relationship with each category. Besides, the PLE consults the neighborhood label information when assigning PLs to avoid near-boundary ambiguous PLs, which cannot be fulfilled by traditional isolated PL assignment strategies. By jointly considering the network prediction, the geometric property, and the neighborhood information, the PLE can boost pseudo-label accuracy and, hence, lead to positive intrusion knowledge transfer.

The GGA then formulates each domain as a graph, where vertices and edges represent intrusion categories and their inter-relationships. As illustrated in Fig. 2, enforcing a perfect

GGA can have each intrusion category and their inter-relationships well aligned between domains. First, with the help of the PLE, the GGA performs a graph-level shape-keeping via a confused discriminator which is incapable to distinguish weighted adjacency matrices (WAMs) between intrusion domain graphs. Upon aligning the graph shapes, a center point matching mechanism and a rotation avoidance mechanism avoid graph misalignment caused by symmetry and graph rotation, respectively. Finally, the GGA will perform a vertex-level matching by preserving categorical correlation knowledge between domains, which equivalently aligns the graph vertex of each intrusion category between domains. Holistically, they form a graph alignment framework from general to specific level from a geometric perspective. The GGA can robustly transfer the enriched intrusion knowledge from the NI domain to facilitate more accurate intrusion detection in the II domain and, hence, secure IoT infrastructures. GGA's motivation has been illustrated in Figs. 1 and 2.

In summary, the contributions of this article are threefold.

- 1) We propose to transfer enriched intrusion knowledge from the NI domain to facilitate more accurate intrusion detection for data-scarce IoT domains and formulate it as a semi-supervised HDA problem.
- 2) To our best knowledge, we are the pioneer to tackle this HDA problem from a pure GGA perspective with the help of the PLE mechanism. Rather than using isolated coarse-grained PL strategies, the PLE makes fine-grained PL assignment by jointly considering geometric and neighborhood information to filter confident but geometrically incorrect PLs and near-boundary ambiguous PLs. The GGA then aligns domain graphs geometrically through four mechanisms, holistically forming a graph alignment framework from graph to vertex granularity.
- 3) We conduct comprehensive experiments of several tasks on five widely used intrusion detection data sets to verify the superior performance achieved by the GGA, and show the usefulness of its constituting components.

The remainder of this article is organized as follows. Section II presents related works by categories and demonstrates the research opportunities of the GGA method. Section III provides model preliminaries, graph formulations, and the GGA model architecture. The detailed GGA and pseudo-label election mechanism are explained in Section IV. Section V presents experimental setups and result analyzes. The last section concludes this article. For better readability, an acronym table, and a notation table have been presented in the Appendix section.

II. RELATED WORK

IID3 Methods: The IID has drawn wide research attention to secure IoT infrastructures. The rule-based IID methods were initially popular. Dietz et al. [3] performed automatic scanning of neighboring IoT devices for known vulnerability patterns and temporarily isolated detected compromised devices. Jun and Chi [12] proposed to filter security violations via complex event processing, which required a sophisticated rule repository. On the other hand, machine learning

techniques were also widely used for IID. Anthi et al. [13] presented a three-layer intrusion detector that worked under a supervised fashion for smart home settings. Shukla [14] tackled the problem via a hybrid two-stage mechanism that combined K -means clustering and decision trees. On the deep learning perspective, models, such as feedforward neural network, deep autoencoder, and BiLSTM RNN, were utilized to work on the IID problem by [4], [15], and [16], respectively. However, these methods either needed a sophisticated intrusion pattern repository, which requires substantial expertise to build and can hardly be complete and up-to-date or required abundant amount of fully labeled data for training, which is labor-intensive to annotate. Hence, it naturally leads to the DA-based methods which can comfortably work under challenging data-scarce IID scenarios.

DA and Intrusion Detection: DA leverages source domain knowledge to facilitate better learning for data-scarce target domains, and, hence, is suitable to tackle the intrusion detection for the data-scarce IoT domain. Vu et al. [17] utilized two autoencoders on source and target data set and forced the alignment of the bottleneck layers. Later, Ning [5] leveraged the Laddernet to tackle IID under a semi-supervised setting. Hu et al. [6] presented a deep subdomain adaptation network with attention mechanism that performed intrusion knowledge transfer by minimizing local maximum mean discrepancy. However, these methods did not use PL assignments to exploit unlabeled target domain and failed to perform DA via a geometric graph-based approach, hence, did not preserve geometric properties during intrusion knowledge transfer. On the other hand, Chen et al. [18] tackled the intrusion domain alignment problem via transfer neural tree (TNT), a unified framework that combined feature mapping, adaptation, and classification. A generalized joint distribution adaptation (G-JDA) approach was presented [9] to learn a pair of feature projectors to eliminate the marginal and conditional distribution divergence. Yao et al. [19] proposed the DDA method that applied an adaptive classifier to reduce distribution divergence and enlarge interclass divergence. The TNT, G-JDA, and DDA applied direct prediction as PLs for unlabeled target instances and completely ignored the label neighborhood information. Yao et al. [11] put forward the STN, a conditional distribution alignment strategy with the help of a soft-label paradigm. Singh et al. [20] presented the STAR framework, which emphasized unlabeled target instances based on the distance with closest class prototypes during intrusion domain alignment. Saito et al. [21] achieved intrusion knowledge transfer by optimizing the minimax loss on the domain conditional entropy (MME). It utilized unlabeled target data based on a threshold-based strategy. The APE [22] method chose to alleviate intradomain discrepancy via three procedures, namely, Attraction, Perturbation, and Exploration. From a clustering-based perspective, Li et al. [10] proposed cross-domain adaptive clustering (CDAC) to tackle the DA problem. In MME, APE, and CDAC, unlabeled target instances will be pseudo-labeled based on a threshold strategy. However, when assigning PLs, these method either failed to jointly consider geometric properties or assigned PLs in an isolated manner that ignored the relationships between pseudo-labeled

instances and their neighboring labeled instances, compromising accurate intrusion knowledge transfer. Some methods also required a manual threshold set based on prior experience and was not generalizable between tasks.

Tackling intrusion domain alignment from a graph perspective is also feasible. For example, the WCGN method matched domains via graph learning [7], [8] to benefit the domain alignment. Pilanci and Vural [23] proposed a graph base alignment method by transferring the graph spectrum information. However, although these embedding-related methods can partially convey the geometric information of domains, learning a good embedding is highly data-dependent, hindering their applicability. Besides, none of these graph-based methods solved the graph matching problem from a pure GGA perspective, which left a void to be filled.

Our method tackles the HDA problem from a pure geometric graph perspective, which jointly considers several levels of geometric property matching. The GGA method does not require a huge amount of data for graph embedding learning and enjoys a relatively low complexity. Besides, we utilize a PLE mechanism which jointly accounts for the geometric properties and the neighborhood information, so that the confident but wrong PL prediction that violates geometric properties and near-boundary ambiguous PLs can be avoided for positive transfer. Finally, we utilize the GGA method to facilitate more accurate intrusion detection for the data-scarce IoT domain.

III. MODEL PRELIMINARY AND ARCHITECTURE

A. Model Preliminary

The GGA method works under a semi-supervised HDA setting. It involves a source NI domain formulated as follows:

$$\mathcal{D}_S = \{\mathcal{X}_S, \mathcal{Y}_S\} = \{(x_{Si}, y_{Si})\}, i \in [1, n_S]$$

$$x_{Si} \in \mathbb{R}^{d_S}, y_{Si} \in [1, K] \quad (1)$$

where the source NI domain contains n_S traffic records with their corresponding intrusion label. Each record is represented using d_S features, and there are K categories. Similarly, the target II domain are defined as follows:

$$\mathcal{D}_{TL} = \{\mathcal{X}_{TL}, \mathcal{Y}_{TL}\} = \{(x_{TLi}, y_{TLi})\}, i \in [1, n_{TL}]$$

$$\mathcal{D}_{TU} = \{\mathcal{X}_{TU}\} = \{(x_{TUj})\}, j \in [1, n_{TU}]$$

$$\mathcal{D}_T = \mathcal{D}_{TL} \cup \mathcal{D}_{TU}, x_{TLi}, x_{TUj} \in \mathbb{R}^{d_T}, y_{TLi} \in [1, K]$$

$$n_T = n_{TL} + n_{TU}, n_{TL} \ll n_{TU}. \quad (2)$$

Under the semi-supervised setting, the target domain is scarcely labeled, i.e., $n_{TL} \ll n_{TU}$. The source and target domain present heterogeneities, such as belonging to different feature spaces, i.e., $d_S \neq d_T$.

B. Graph Formulation

To perform the GGA, we formulate each intrusion domain as a graph, i.e., $G_X = \langle V_X, E_X \rangle$, $X \in \{S, T\}$. Both domains share K intrusion categories, therefore, each graph has K vertices, the vertex V_i is the centroid of the category i , denoted

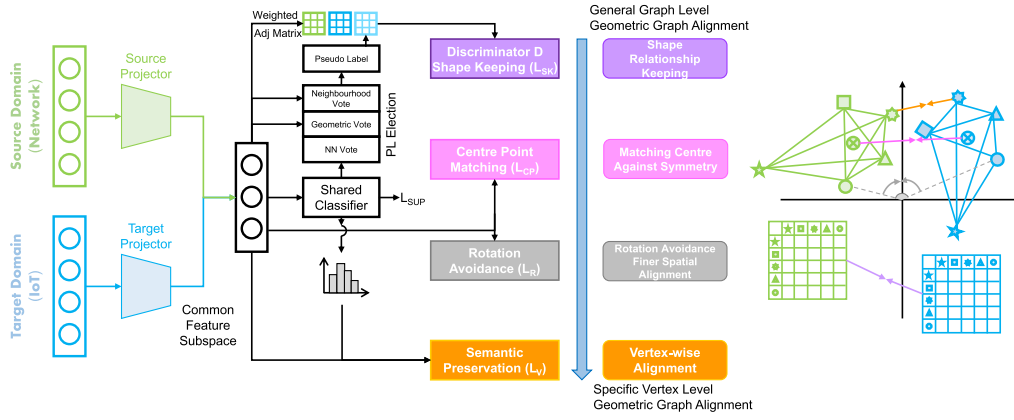


Fig. 3. Architecture of the GGA method.

as follows:

$$V_X^i = \frac{1}{n_X^i} \sum_{j=1}^{n_X^i} x_{X_j}, i \in [1, K], X \in \{S, T\} \quad (3)$$

where n^i is the number of records under category i . Both graphs are formulated as a complete graph, the weight of edge $E < V_X^i, V_X^j >$ is set to be the Euclidean distance between vertex V_X^i and V_X^j . The corresponding WAMs are denoted as M_S and M_T .

C. Model Architecture and Overview

The architecture of the GGA method has been shown in Fig. 3. Each intrusion domain has a feature projector that maps features into a common feature subspace with dimension d_C . The feature projector is defined as follows:

$$f(x_i) = \begin{cases} E_S(x_i), & \text{if } x_i \in \mathcal{X}_S \\ E_T(x_i), & \text{if } x_i \in \mathcal{X}_T = \mathcal{X}_{TL} \cup \mathcal{X}_{TU} \end{cases} \quad (4)$$

$$f(x_i) \in \mathbb{R}^{d_C}.$$

The GGA method will then utilize the PLE mechanism to assign fine-grained PLs to unlabeled target data and avoid error-prone PLs to mitigate negative transfer. To perform GGA between these heterogeneous domains, the WAM of the source data, the labeled target data, and the combination of labeled and pseudo-labeled target data will be generated to confuse the discriminator D . Highly similar WAMs indicate well-aligned intrusion categories and the fine preservation of category-wise interrelations and is equivalent with a perfect geometric graph shape keeping. By fusing three WAMs, the geometric shape of domain graphs are aligned, meanwhile, the labeled and pseudo-labeled target data will be better fused together. After keeping the shape, the domain graphs can still misalign due to rotation and symmetry, which are mitigated by the rotation avoidance mechanism and the center point matching. Besides, categorical correlations yielded by the shared classifier C will be preserved between domains, which acts as a vertex-level alignment mechanism. Holistically, the GGA approach aligns the domain graphs from general shape level to specific vertex

level. The motivation is to align the domain graphs in a fine-grained manner, such that the shared classifier C yields the best intrusion detection accuracy for unlabeled target domain.

IV. GGA ALGORITHM

In this section, we will first introduce the PLE mechanism which facilitates better target participation during the GGA process. Then, the GGA process and its constituting components will be explained.

A. Pseudo-Label Election Mechanism

Assigning PLs to unlabeled target data can excavate its potentials during intrusion knowledge transfer. However, erroneous PL assignment may mislead the model toward negative transfer. Traditional efforts mainly assigned PL to instances in an isolated manner, without considering the relationship between labels, and suffered from issues such as confident but geometrically-inconsistent PLs and near-support ambiguous PLs. Therefore, we utilize the PLE mechanism to mitigate these issues as much as possible. The PLE jointly considers the voting of NN prediction, the geometric properties, and neighborhood information. A PL assignment will only be made for an instance if these three votes reach a consensus. When producing the geometric property-based PL, the category of the most Cosine-similar labeled data centroid μ_{S+TL}^i will be utilized as PL_G for each unlabeled target instance and is defined as follows:

$$PL_G^i = \underset{k}{\operatorname{argmax}} \operatorname{COS}(\mu_{S+TL}^{(k)}, x_{TU}^i) \quad (5)$$

$$\mu_{S+TL}^{(k)} = \frac{1}{n_S^{(k)} + n_{TL}^{(k)}} \sum_{x_i \in \mathcal{X}_S^{(k)} \cup \mathcal{X}_{TL}^{(k)}} x_i$$

where PL_G^i denotes the geometric-based PL for the x_{TU}^i , $\mathcal{X}_S^{(k)}$ denotes source instances from category k . If the NN-prediction yields a confident PL prediction but is inconsistent with the geometric similarity property, then such confident but contradictory PL will be rejected, as illustrated in Fig. 4(b). Besides, the PLE will also consult the neighborhood information when assigning PLs. If the K -nearest neighborhood around an unlabeled target instance cannot reach a majority agreement or

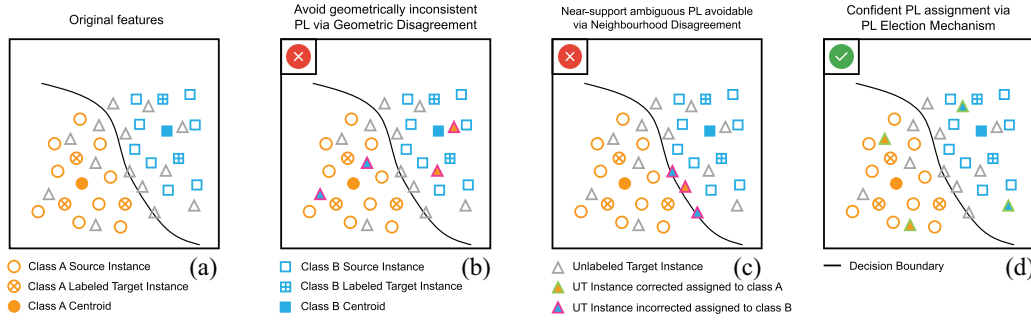


Fig. 4. Illustration of the PLE. (a) Original feature. (b) Avoid geometrically inconsistent PL via geometric disagreement. (c) Near-boundary ambiguous PL assignment avoidable via neighborhood disagreement. (d) PLE's assignment.

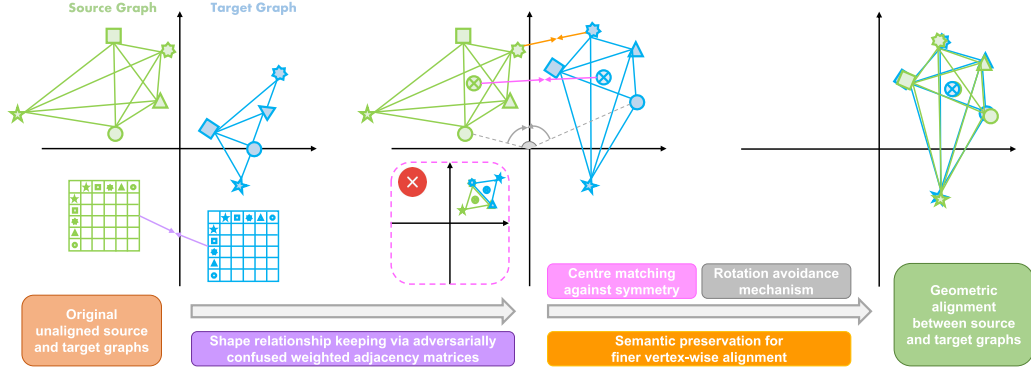


Fig. 5. Illustration of the GGA method.

reach an agreement against the NN prediction or the geometric vote, then such assignment will also be rejected. This is useful, especially when deciding the PL for near-boundary unlabeled instances, as illustrated in Fig. 4(c). Since the neighborhood can be harder to reach an agreement near the boundary due to ambiguity, the PLE can effectively get rid of near-boundary PLs which are more likely to be erroneous. Overall, the PLE will only assign confident PLs with probabilistic, geometric, and neighborhood soundness, which can significantly boost the PL accuracy and, therefore, lead to positive intrusion knowledge transfer.

B. Geometric Graph Alignment

The GGA method has been illustrated in Fig. 5. It will perform GGA from the general graph granularity to the specific vertex granularity, i.e., the shape keeping via confused discriminator (purple step), rotation avoidance mechanism (gray step), center point matching against symmetry (pink step), and the vertex-level alignment via semantic preservation (orange step).

Shape Keeping: First, the GGA will align the graph shape via a confused discriminator. We define the *Same Shape Rule* as follows.

Definition 1 (Same Shape Rule): Both G_S and G_T have their shape aligned with each other if and only if the WAMs M_S and M_T are the same.

Specifically, with the help from the PLE, the GGA will construct three WAMs: 1) the source WAM M_S ; 2) the labeled target WAM M_{TL} ; and 3) a WAM based on both labeled and

pseudo-labeled target data M_{TL+PL} . These WAMs will then be flattened and feed into the discriminator D , a single-layer neural network that tries to distinguish the origin of the input WAM. The source domain WAM M_S is assigned with domain label 1, while target domain WAMs are assigned with domain label 0. The shape keeping loss $\mathcal{L}_{SK}$ is defined as follows:

$$\mathcal{L}_{SK} = \log(D(M_S)) + \frac{1}{2} \sum_{M \in \{M_{TL}, M_{TL+PL}\}} (1 - \log(D(M))). \quad (6)$$

The source and target projector will try to minimize the $\mathcal{L}_{SK}$ and let the discriminator D to be unable to distinguish the origin of the input WAMs, while the discriminator will try to stay unconfused. Upon this minimax game reaches an equilibrium, both G_S and G_T will have their shape aligned as indicated in Fig. 5 (Mid), and the labeled and pseudo-labeled target data will be better fused together.

Rotation Avoidance Against Rotated Misalignment: As indicated in Fig. 5 (Mid), graph rotation can still cause domain graph misalignment, even though the shape is aligned. Therefore, to further align the domain graphs geometrically, we define the *Same Angle Rule* as follows:

Definition 2 (Same Angle Rule): For graph G_S and G_T , the Same Angle Rule holds if $\forall i \in [1, K], 1 - \cos(V_S^i, V_T^i) = 0$.

The GGA method will keep the Same Angle Rule holding by minimizing the rotation loss, defined as follows:

$$\mathcal{L}_R = \sum_{i=1}^K (1 - \cos(V_S^i, V_T^i)). \quad (7)$$

By minimizing $\mathcal{L}_R$, graph misalignment caused by rotation will be prevented since the categorize-wise vertex angle is enforced to be 0° .

Centre Point Matching Against Symmetric Misalignment: As shown in the pink-boxed example in Fig. 5 (Mid), upon fixing the graph shape and enforcing the Same Angle Rule, symmetry can still cause misalignment. Therefore, we further define the *Same Centre Rule* as follows:

Definition 3 (Same Centre Rule): For graph G_S and G_T , the Same Centre Rule holds if $\mu_S = \mu_T$, $\mu_X = (1/n_X) \sum_{i=1}^{n_X} \mathcal{X}_{X_i}$, $X \in \{S, T\}$.

We minimize the centre point matching loss as follows:

$$\mathcal{L}_{CP} = \|\mu_S - \mu_T\|^2. \quad (8)$$

Enforcing the Same Centre Rule brings threefold advantages: First, the graph misalignment caused by symmetry can be prevented; Then, it boosts data participation during the intrusion knowledge transfer to fully excavate the potentials in all data, irrespective of whether a PL is assigned or not; Finally, despite its computational simplicity, it can boost the intrusion detection performance as indicated by the experiments.

Vertex-Level Alignment via Semantic Preservation: The above three steps focus on the overall graph level, in this step, we shift our focus to the vertex granularity. Each vertex in the domain graph represents an intrusion category centroid, during prediction, it presents a unique probabilistic category-wise correlation. Use object as an example, a laptop should be highly similar with other laptops, somewhat similar with a TV screen, and very dissimilar with a bike, irrespective of its domain origin. By enforcing the corresponding vertices from both domain graphs to preserve the correlation semantic, it in turn forces vertex-level alignment between domain graphs. Specifically, for source category k , its correlation semantic is defined as follows:

$$q^{(k)} = \frac{1}{n_S^{(k)}} \sum_{x_i \in \mathcal{X}_S^{(k)}} \text{softmax}\left(\frac{C(f(x_i))}{T}\right) \quad (9)$$

where $\mathcal{X}_S^{(i)}$ denotes category i source instances, C and f denote the shared classifier and the feature projector, respectively, T is a temperature hyperparameter that controls the correlation semantic smoothness. Similarly, the correlation semantic of each labeled target instance is defined as follows:

$$p_i = \text{softmax}(C(f(x_i))), x_i \in \mathcal{X}_{TL}. \quad (10)$$

To perform the semantic preservation, we minimize the vertex-level alignment loss as follows:

$$\mathcal{L}_{VS} = -\frac{1}{n_{TL}} \sum_{(x_i, y_i) \in (\mathcal{X}_{TL}, \mathcal{Y}_{TL})} \left(q^{(y_i)}\right)^\top \log(p_i). \quad (11)$$

Together with the supervision provided by the labeled target instances, the final vertex-level alignment loss is defined as follows:

$$\mathcal{L}_V = \frac{1-\alpha}{n_{TL}} \sum_{(x_i, y_i) \in (\mathcal{X}_{TL}, \mathcal{Y}_{TL})} \mathcal{L}_{ce}(C(f(x_i)), y_i) + \alpha \mathcal{L}_{VS} \quad (12)$$

where $\mathcal{L}_{ce}$ denotes cross entropy loss. By minimizing the vertex-level alignment loss $\mathcal{L}_V$, it will enforce vertices in the same category from different graphs to align with each other. Otherwise, the correlation semantic will fail to be preserved.

GGA Theorem: The GGA forms the above mechanism into a holistic framework and can align two domain graphs with theoretical guarantee. We state the *GGA Theorem* as follows.

Theorem 1 (GGA Theorem): Given graph G_S and G_T , if the Same Shape Rule, the Same Angle Rule, and the Same Centre Rule hold simultaneously, then G_S and G_T must exactly align with each other.

The proof of the GGA Theorem is as follows.

Proof: We prove the GGA Theorem by induction with the help of contradiction.

Case 1: Both G_S and G_T have two vertices. Given that the Same Angle Rule holds, both graphs must be parallel with each other. Given that the Same Shape Rule holds, it enforces the only edge in both graphs to have equal length. Therefore, it is trivial to conclude that these two graphs are the same.

Case k: Both G_S and G_T have k vertices. The aforementioned three rules hold and G_S and G_T are aligned. We add an additional vertex to G_S and G_T separately. Without breaking any of the aforementioned three rules, the new graph G'_S and G'_T also align with each other.

Proof Case k by Contradiction: Since under Case k, the pre-requisite states that G_S and G_T align with each other, therefore, we simply denote both of them as G_X .

Case k1: If G_X is asymmetric regarding any line, then it is trivial that there is no possible strategy to add point differently to G_X to get G'_X and G''_X without violating the Same Centre Rule.

Case k2: If G_X is symmetric regarding a symmetric axis, to stick to the Same Shape Rule and the Same Angle Rule, the only possible strategy to add V_S^{k+1} and V_T^{k+1} is as follows: the line crossing V_S^{k+1} and V_T^{k+1} should also cross-origin (Same Angle Rule) and the center point of the symmetric axis (Same Shape Rule). However, if we add V_S^{k+1} and V_T^{k+1} differently, then they must be symmetric regarding the graph symmetric axis, which violates the Same Centre Rule for the graphs. ■

Given that the GGA theorem holds, the constituting components of the GGA method can achieve a fine-grained GGA with theoretical guarantee and benefit intrusion knowledge transfer.

C. Overall Optimization Objective

Finally, the source labels provide supervision during the training process with supervision loss defined as follows:

$$\mathcal{L}_{SUP}(\mathcal{X}_S, \mathcal{Y}_S) = \frac{1}{n_S} \sum_{(x_i, y_i) \in (\mathcal{X}_S, \mathcal{Y}_S)} \mathcal{L}_{ce}(c(f(x_i)), y_i). \quad (13)$$

Overall, the optimization objective of GGA is as follows:

$$\min_{C, E_S, E_T} \max_D (\mathcal{L}_{SUP} + \gamma \mathcal{L}_{SK} + \eta \mathcal{L}_R + \lambda \mathcal{L}_{CP} + \mathcal{L}_V) \quad (14)$$

where γ , η , and λ are hyperparameters controlling the influence of loss components during optimization. During initial training stages, both domain graphs may suffer from immature shape, therefore, the γ is set to a relatively low value

to emphasize other components such as vertex-wise semantic alignment, etc. As the training progresses, the γ will grow linearly to gradually emphasize the importance of shape keeping. To form the optimization into an end-to-end procedure, we follow [24] to apply the gradient reversal layer for the discriminator and optimize the model using the Adam gradient descent. Upon the equilibrium of the above minimax game is reached, the network training concludes, and the domain graphs can be aligned in a fine-grained manner, which facilitates more accurate intrusion detection for the target IoT domain.

V. EXPERIMENT

A. Experimental Data Sets

During experiments, we utilize five representative and comprehensive intrusion detection data sets, which include three NI data sets: 1) NSL-KDD; 2) UNSW-NB15; and 3) CICIDS2017, and two II data sets: 1) UNSW-BOTIOT and 2) UNSW-TONIOT.

NI Data Set: NSL-KDD The NSL-KDD data set [25] was released in 2009, which addressed issues of the prior data set KDD CUP99 [26] such as having lots of redundant records. The NSL-KDD data set contains benign traffic with four types of intrusions, such as probing attack, Denial-of-Service (DoS) attack, etc. Following [13], we reasonably utilize 20% of the data set. Each traffic record in the data set is represented using 41 features. We follow Harb et al. [27] to use the top 31 most informative features as the feature representation. The data set is denoted as K .

NI Data Set: UNSW-NB15 The UNSW-NB15 data set [28] was created by UNSW in 2015 using the IXIA PerfectStorm tool, which aimed to address the data quality issue and out-of-date incomprehensive network flow issue observed in previous data sets. The data set contains ten traffic categories, including normal traffic, DoS attacks, reconnaissance attacks, etc. We utilize 2700 traffic records, which follows the data set magnitude in [29]. The data set is represented using 49 features, we perform preprocessing to remove four features having value 0 for nearly all records. The data set is denoted as N .

NI Data Set: CICIDS2017 The CICIDS2017 data set [30] was released in 2017, which served as one of the most up-to-date NI data sets with modern attack patterns. The data set has seven types of intrusions with benign traffic, represented in 77-D features. We utilize the 20% portion of the data set provided by the data set creator to perform the model training and testing. During preprocessing, we perform data deduplication and categorical-numerical entries conversion. Following Stiawan et al. [31], we use features with top 40 information gain as the feature space of the data set and denote the data set as C .

II Data Set (UNSW-BOTIOT) The UNSW-BOTIOT data set [32] was released in 2017 by UNSW, which presented up-to-date modern attack scenarios captured based on a realistic testbed environment. The testbed environment deployed IoT devices, such as weather station, smart fridges, etc., and utilized MQTT protocol, a lightweight IoT communication protocol commonly used in realistic IoT scenarios. The data set quality has been carefully addressed, and the attack

diversity has been improved. The data set contains four categories, including normal traffic, DoS attacks, information theft attacks, etc. Following [29], we utilize 10 000 data records. The original data set utilizes a 46-D feature space. We follow the data set creator's advice to use top 10 most informative features as the feature space. The data set is denoted as B .

II Data Set (UNSW-TONIOT) The UNSW-TONIOT data set [33] serves as one of the latest II data sets [34], released in 2021. It reflects modern IoT standards, protocols, and is captured on modern testbed consists of seven types of IoT devices, such as smart fridge, modbus sensor, GPS tracker, etc. The data set covers seven types of intrusions, including scanning attacks, DoS attacks, etc. Heterogeneities present between IoT devices as the features captured by each type of IoT device have their own feature dimension. Following [29], [35], we utilize around 10% of the data set, and select the weather meter and GPS tracker as the IoT devices used during experiments, which are denoted as W and G , respectively.

Comprehensiveness of Data Sets The data sets we utilized are representative and comprehensive to verify the effectiveness of the proposed method. First, these data sets are widely recognized and widely adopted by the research community to testify intrusion detection effectiveness with thousands of citations. Second, these data sets are developed and released in recent years, some of them are release in 2021, therefore, they can reflect current intrusion trends and methods. Finally, these data sets are captured on realistic testbeds with large-scale real IoT devices, and the sufficiency of the testbed is recognized by the research community. Hence, these data sets are representative with guaranteed comprehensiveness.

B. Shared Intrusion

The NI data sets and the II data sets can have at most eight shared categories that can be transferred as intrusion knowledge, such as DoS attack, password attack, backdoor attack, etc. These shared intrusion categories account for 100%, 54.9%, 100%, 100%, and 98.3% the amount of records in NSL-KDD, UNSW-NB15, CICIDS2017, UNSW-BOTIOT, and UNSW-TONIOT data set, respectively. Therefore, transferring intrusion knowledge with wide coverage can effectively detect most modern intrusions targeting the IoT domain.

C. Implementation Details

We implement the GGA method using the PyTorch deep learning framework. Following [11], [36], feature projectors are two-layer neural networks using LeakyReLU [37] as their activation function. Both the shared classifier and the discriminator are implemented as single-layer neural networks. The hyperparameter setting is fixed during all experiments as follows: $\alpha = 0.1$, $\gamma_{\min} = 0.01$, $\gamma_{\max} = 0.1$, $\eta = 0.01$, $\lambda = 0.01$, $d_C = 3$, $T = 5$, and $\#neighbour = 4$. Note that γ will increase linearly from $\gamma_{\min}$ to $\gamma_{\max}$ as the training progresses. To emphasize the target data scarcity, we set $n_{TL} : n_{TU} = 1 : 50$ as the default ratio. We also conduct the parameter sensitivity analyses to verify the stable and robust performance of the GGA method. Following [29], [38], we

TABLE I
INTRUSION DETECTION ACCURACY UNDER DEFAULT DATA SCARCITY RATIO $n_{TL} : n_{TU} = 1 : 50$

S $\rightarrow$ T	C $\rightarrow$ B	N $\rightarrow$ B	K $\rightarrow$ B	C $\rightarrow$ G	N $\rightarrow$ G	K $\rightarrow$ G	C $\rightarrow$ W	N $\rightarrow$ W	K $\rightarrow$ W	Avg
TNT	10.71	16.42	16.40	53.85	70.07	61.37	53.41	70.07	61.36	45.96
APE	33.99	51.46	52.42	54.18	70.35	61.67	47.24	63.35	55.81	54.50
MME	34.98	51.35	52.67	54.14	70.29	61.63	50.44	65.46	57.81	55.42
CDAC	34.04	50.29	50.87	54.22	70.35	61.70	53.89	70.34	61.65	56.37
STAR	33.69	50.87	51.02	54.17	70.32	61.65	53.72	70.23	61.83	56.39
DDAS	35.13	52.15	52.01	53.90	72.52	60.11	54.35	71.54	63.48	57.24
STN	35.93	51.94	53.48	57.44	69.85	64.84	54.28	71.28	63.24	58.03
DDAC	35.20	52.54	53.55	55.65	71.92	59.36	56.96	71.80	65.34	58.04
WCGN	42.79	60.10	63.80	58.26	76.62	66.83	56.82	72.39	64.41	62.45
GGA (Ours)	48.59	68.99	77.26	59.43	79.18	68.25	58.71	72.75	66.22	66.60

use unlabeled target prediction accuracy as our major evaluation metric, and also use the category-weighted precision (P), recall (R), F1-score (F), and Area under the ROC Curve (A) [39], [40] to evaluate the GGA performance. Specifically, we define true positive $TP^{(k)}$ as the number of category k intrusions being correctly identified, similar for true negative $TN^{(k)}$, false positive $FP^{(k)}$, and false negative $FN^{(k)}$. The mathematical definitions of evaluation metrics are as follows:

$$\text{Accuracy} = \frac{\sum_{k=1}^K (TP^{(k)} + TN^{(k)})}{n_{TU}} \quad (15)$$

$$\begin{aligned} \text{Precision} &= \sum_{k=1}^K \frac{|\mathcal{X}_{TU}^{(k)}|}{n_{TU}} \cdot \frac{TP^{(k)}}{TP^{(k)} + FP^{(k)}} \\ &= \sum_{k=1}^K \frac{|\mathcal{X}_{TU}^{(k)}|}{n_{TU}} \cdot \text{Precision}^{(k)} \end{aligned} \quad (16)$$

$$\begin{aligned} \text{Recall} &= \sum_{k=1}^K \frac{|\mathcal{X}_{TU}^{(k)}|}{n_{TU}} \cdot \frac{TP^{(k)}}{TP^{(k)} + FN^{(k)}} \\ &= \sum_{k=1}^K \frac{|\mathcal{X}_{TU}^{(k)}|}{n_{TU}} \cdot \text{Recall}^{(k)} \end{aligned} \quad (17)$$

$$F1 = \sum_{k=1}^K \frac{|\mathcal{X}_{TU}^{(k)}|}{n_{TU}} \cdot \frac{2 \cdot \text{Precision}^{(k)} \cdot \text{Recall}^{(k)}}{\text{Precision}^{(k)} + \text{Recall}^{(k)}} \quad (18)$$

Besides, metrics A (AUC) represents the area under the ROC curve, a curve plotting the TP rate and the FP rate.

D. State-of-the-Art Baselines

We utilize nine state-of-the-art comparing methods, including TNT [18], MME [21], STN [11], APE [22], DDAS, DDAC [19], WCGN [7], [8], CDAC [10], and STAR [20]. All of them are from top-tier conferences and journals, and eight of them are proposed between 2019 and 2021. We summarize their differences with GGA as follows.

- 1) From the pseudo-labeling perspective, the DDAC, DDAS, WCGN, and TNT utilize predicted hard pseudo label for target instances and ignore both the geometric property and neighborhood information. On the other hand, MME, CDAC, and APE involve threshold-based pseudo-label strategy. However, setting thresholds properly requires expertise experience and has compromised flexibility. Both APE and STAR apply

pseudo-label strategy using geometric property as a reference, however, the neighborhood information is still ignored. Besides, STN utilizes a soft-label strategy, lacking emphasize on confident predictions. The GGA is the only method that jointly considers both the geometric property and neighborhood information, while avoiding hard-to-set threshold.

- 2) From the domain alignment perspective, these baselines apply diverse techniques, such as CDAC's adversarial adaptive clustering, MME's alternated conditional entropy minimization, STN's joint distribution matching, etc. However, these methods fail to explicitly conduct domain alignment from a geometric graph perspective. To our best knowledge, there lacks a similar pure geometric-based graph baseline method, the WCGN is a comparable state-of-the-art graph method based on graph learning framework. However, since a proper graph learning requires both sufficient data and a high complexity, it is challenging under the data-scarce and computationally constrained IoT scenario. Conversely, the GGA performs knowledge transfer via a GGA perspective. It fills the void of previous methods, avoids heavy data dependency, and has a relatively low complexity.

Therefore, these state-of-the-art baseline methods are representative and can be used to verify the superiority of the GGA method from different perspectives.

E. Performance Evaluation

Performance Analysis Under Default Data Scarcity Ratio:

We analyze the performance against state-of-the-art counterparts under the default target data scarcity ratio. As indicated in Table I, the GGA clearly outperforms all comparing methods by at least 4.2%. The best-performed comparing method WCGN utilizes graph learning framework, however, it does not perform well under the data-scarce IoT scenario, its PL assignment strategy also lacks consideration of geometric properties and assigns PL in an isolated manner. Hence, it is natural to observe a performance boost achieved by the GGA.

Performance Analysis Under Diverse Data Scarcity Ratios:

To verify the effectiveness and robustness of the GGA under varied target data scarcities, we vary the data scarcity ratio $n_{TL} : n_{TU}$ between 1 : 10 and 1 : 100. Following [5], [11], [36],

TABLE II
INTRUSION DETECTION ACCURACY UNDER VARIED $n_{TL} : n_{TU}$ RATIOS

S → T	N → B			K → G			C → W			Overall	1 : 100
$n_{TL} : n_{TU}$	1 : 10	1 : 50	1 : 100	1 : 10	1 : 50	1 : 100	1 : 10	1 : 50	1 : 100	Avg	Avg
TNT	18.86	16.42	16.38	61.40	61.37	61.30	53.44	53.41	53.31	43.99	43.66
APE	51.84	51.46	47.14	61.70	61.67	61.50	53.75	47.24	31.76	52.01	46.80
MME	52.84	51.35	47.01	61.65	61.63	45.23	53.36	50.44	50.07	52.62	47.44
CDAC	50.70	50.29	50.03	61.73	61.70	61.32	54.24	53.89	53.64	55.28	55.00
STAR	51.96	50.87	50.75	61.70	61.65	61.40	53.75	53.72	53.70	55.50	55.28
DDAS	52.50	52.15	51.64	60.37	60.11	59.70	54.56	54.35	53.71	55.45	55.02
STN	53.71	51.94	51.65	65.15	64.84	61.03	58.68	54.28	54.05	57.26	55.58
DDAC	53.16	52.54	52.25	59.93	59.36	58.61	58.17	56.96	56.60	56.40	55.82
WCGN	61.98	60.10	58.47	67.09	66.83	65.92	58.37	56.82	55.17	61.19	59.85
GGA (Ours)	72.87	68.99	68.26	68.73	68.25	67.79	59.04	58.71	57.30	65.55	64.45

TABLE III
INTRUSION DETECTION PERFORMANCE USING VARIOUS EVALUATION METRICS UNDER DEFAULT DATA SCARCITY

S → T	N → G				K → B			
Metrics	P	R	F	A	P	R	F	A
APE	0.495	0.703	0.581	0.500	0.395	0.516	0.366	0.557
CDAC	0.493	0.702	0.579	0.500	0.251	0.501	0.335	0.553
STAR	0.495	0.703	0.581	0.491	0.250	0.500	0.333	0.481
DDAS	0.700	0.718	0.688	0.696	0.390	0.522	0.376	0.674
STN	0.495	0.703	0.581	0.693	0.525	0.534	0.405	0.768
DDAC	0.749	0.712	0.713	0.700	0.393	0.540	0.407	0.769
WCGN	0.763	0.747	0.740	0.884	0.608	0.617	0.583	0.743
GGA (Ours)	0.828	0.800	0.790	0.927	0.778	0.773	0.757	0.884

the 1 : 100 case is enough to represent an extreme data-scarce setting. We randomly pick three tasks and present their performance in Table II. As we can observe, the GGA achieves the best intrusion detection performance among all three tasks under all data scarcity settings. It yields a 4.36% and 8.29% overall average performance increase compared with the best and second best-performed methods WCGN and STN. Moreover, under the extreme data scarcity case, the performance boost achieved by GGA reaches 4.6% and 8.63% compared with the best and second best-performed counterpart WCGN and DDAC, and only presents a 0.87% drop compared with the performance under the 1 : 50 case, which further verifies the superiority and robustness of the GGA when working under diverse data scarcity conditions.

Performance Analysis Using Diverse Evaluation Metrics: To further verify the effectiveness of the GGA method using evaluation metrics other than accuracy, we randomly select two tasks and record the performance using another four evaluation metrics, and present the result in Table III. We observe GGA achieves superior performance on all evaluation metrics. Specifically, the highest precision indicates that most of the flagged malicious decisions made by GGA are correct. The highest recall means the GGA can flag most amount of intrusions among all malicious traffic. As a harmonic mean of precision and recall, the highest F1-score indicates that the GGA can balance properly between flagging malicious actions and avoiding triggering false alarms. Finally, the highest AUC shows the GGA can effectively distinguish malicious intrusions from normal traffic. Together, these evaluation metrics verify the effectiveness of the GGA method and its real-world usability in terms of false alarm avoidance.

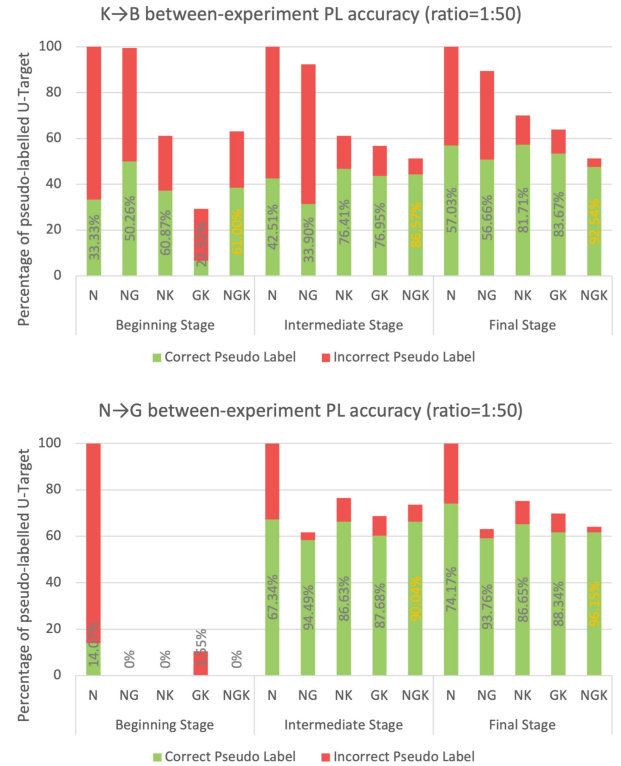


Fig. 6. PL accuracy under default data scarcity ratio between ablated experiments.

Intrusion Detection Performance Summary: We verify the GGA has the best performance on all tasks when evaluated using all metrics. Therefore, it is sufficient to indicate that

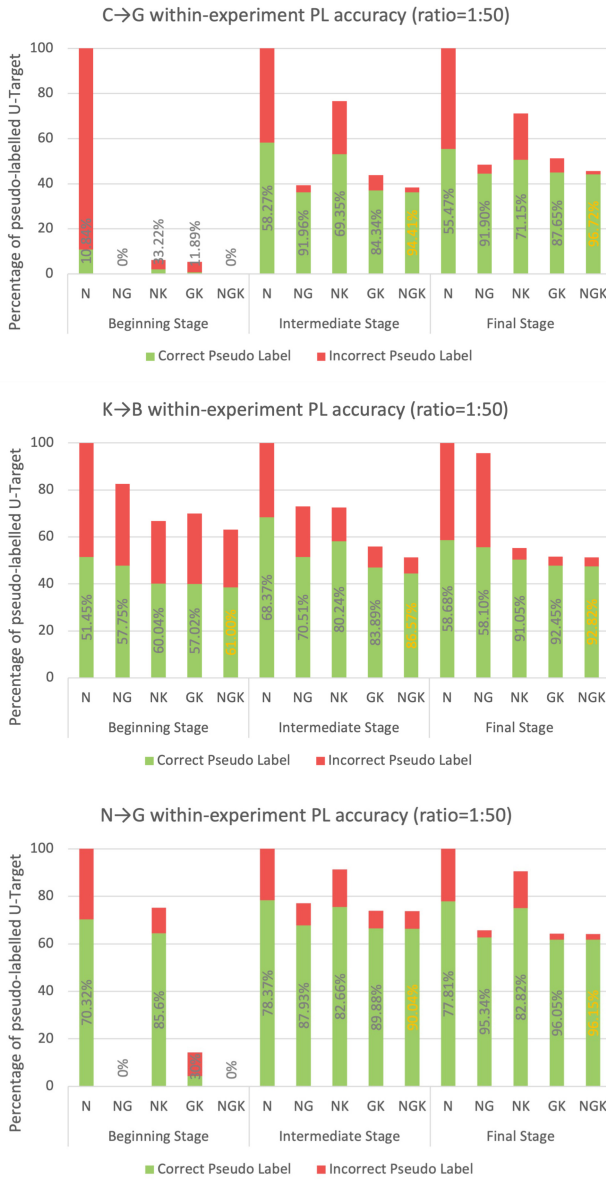


Fig. 7. PL accuracy under default data scarcity ratio within a single experiment that utilizes full PLE.

the GGA method can accurately flag malicious traffic while not causing severe false alarms. The best F1-score and AUC score performance also verify the GGA has the best ability to distinguish benign traffic and different intrusions. Having such capability promotes the real-world usefulness of GGA when performing effective intrusion detection.

F. Pseudo-Label Accuracy Analysis

To justify the efficacy of the PLE mechanism, we perform PL accuracy analysis in three ways: 1) analyze the PL accuracy between ablated experiments under default data scarcity ratio; 2) analyze the PL accuracy of different PLE configurations in a single full PLE setting under default data scarcity ratio; and 3) perform 2) under varied data scarcity ratios to verify the robustness of the PLE.

The results on two randomly selected tasks for case 1) has been illustrated in Fig. 6. Note that N, G, and K represent NN



Fig. 8. Within-experiment PL accuracy under varied data scarcity ratios.

prediction vote, geometric property vote, and neighborhood vote, respectively. The height of each bar represents percentage of unlabeled target data being pseudo-labeled, and red-green color and the value written on each bar indicates the accuracy of PL assignment. As we can observe, using the full PLE yields advantages in threefold.

- 1) The full PLE achieves the highest PL accuracy during all training stages. During the beginning stage, to avoid blindly generating a vast amount of false PLs and mislead the alignment process, the full PLE can even temporarily generate no PLs, since generating PL blindly can only deteriorate the alignment process.
- 2) The full PLE can reach a relatively high PL accuracy around 86.6%–90% even at the intermediate training stage, which guides the aligning process positively by fully exploiting the unlabeled target data.
- 3) The full PLE eventually achieves a PL accuracy around 92.5%–96.2%, which indicates the superiority of PLE on

TABLE IV
ABLATION STUDY GROUP A: INTRUSION DETECTION ACCURACY OF GGA WITH ABLATED GGA COMPONENTS

Experiment	GGA Components				Task			Avg
	Shape Keeping	Rotation Avoidance	Centre Pt Match	Vertex Semantic Match	K → B	N → G	C → W	
A ₁	× ($\gamma_{min}, \gamma_{max} = 0$)	✓	✓	✓	71.83	74.2	56.64	67.56
A ₂	✓	× ($\eta = 0$)	✓	✓	66.75	77.42	56.36	66.84
A ₃	✓	✓	× ($\lambda = 0$)	✓	69.07	75.47	57.74	67.43
A ₄	✓	✓	✓	× ($\alpha = 0$)	75.63	77.77	57.86	70.42
Full GGA	✓	✓	✓	✓	77.26	79.18	58.71	71.72

TABLE V
ABLATION STUDY GROUP B: INTRUSION DETECTION ACCURACY OF GGA WITH ABLATED PLE COMPONENTS

Experiment	Pseudo Label Election Mechanism Components			Task			Avg
	NN Label	Geometric Label	Neighbourhood Label	K → B	N → G	C → W	
B ₁	✓	×	×	72.62	74.37	56.01	67.67
B ₂	✓	✓	×	75.26	75.17	56.72	69.05
B ₃	✓	×	✓	73.76	77.25	57.78	69.60
B ₄	×	✓	✓	69.48	76.75	56.42	67.55
Full GGA	✓	✓	✓	77.26	79.18	58.71	71.72

accurate PL assignment. Although the full PLE may not generate the highest amount of PLs, however, the accuracy matters more than the amount, as indicated by the superior performance achieved by the full PLE during the ablation study.

Besides the PL accuracy analysis performed between ablated experiments, we also perform the PL accuracy analysis within a single full PLE experiment under three randomly picked tasks. As indicated in Fig. 7, the within-experiment performances also comply with the advantages summarized above. The full PLE can stably achieve the highest PL assignment accuracy during all training stages. During each stage, ablating any PLE-constituting component will cause the PL accuracy to drop significantly. This result further verifies the usability of all components considered in the PLE.

To demonstrate the robustness of the PLE under varied data scarcities, the within-experiment PL accuracy is analyzed under varied data scarcity ratios, as indicated in Fig. 8. Under a relatively low-data scarcity case, the PLE can achieve a 99.06% PL accuracy during the final training stage. Even under the extreme data scarcity case, the PL accuracy only drops by 4.27% compared with the 1 : 10 case, which demonstrates that under varied data scarcities, the PLE can work robustly to generate accurate PL assignment and benefit positive intrusion knowledge transfer during the GGA process.

G. Ablation Study

We further investigate the efficacy of GGA's constituting components. Ablation group A has the corresponding GGA components in (14) being turned off. Ablation group B has different PLE voters being ablated. Ablation group C compares GGA with the method that uses the direct vertex Euclidean distance alignment as an alternative, which is defined as follows:

$$\min \sum_{i \in (A,B)} \sum_{i=1}^K \|v_A^i - v_B^i\|^2$$

$$(A, B) \in \{(S, TL + PL), (S, S + TL + PL), (TL + PL, S + TL + PL)\} \quad (19)$$

TABLE VI
ABLATION STUDY GROUP C: INTRUSION DETECTION ACCURACY OF METHODS WITH DIFFERENT GRAPH ALIGNMENT MECHANISMS

Experiment	Alignment Mechanism	Task			Avg
		K→B	N→G	C→W	
C	Vertex E-dist Alignment	74.38	74.79	57.65	68.94
Full GGA	Geometric Graph Alignment	77.26	79.18	58.71	71.72

where $S + TL + PL$ stands for combining instances from the source, labeled-target, and pseudo-labeled target domains.

As indicated in Tables IV–VI, the full GGA outperforms all its ablated counterparts by 3.4% on average, verifying positive contributions made by all constituting components toward a fine-grained GGA. In ablation group A, the rotation avoidance mechanism is the best performance contributor with 4.9% of performance boost, followed by the centre matching, shape-keeping mechanism and vertex-level semantic preservation. In ablation group B, the results verify that all three voting components are indispensable. The performance will drop by 3.3% on average without any one of them. Finally, in ablation group C, a 2.8% performance reduction is observed by the Euclidean distance-based pure vertex alignment procedure. It is natural to observe since there are huge heterogeneities between domains, therefore, pure vertex-level distance-based alignment may not be strong enough to enforce a fine-grained graph alignment, which results in degraded intrusion knowledge transfer. By jointly considering several granularities from shape keeping to vertex-level alignment, the GGA can facilitate a finer alignment and an enhanced intrusion detection performance.

H. Hypothesis Testing for Ablation Study

To verify the statistical significance of the contributions made by each constituting component, i.e., the performance boost is not observed randomly by chance, we perform significance T -test on three randomly selected tasks. As illustrated in Fig. 9, the gray shaded area denotes the significant threshold

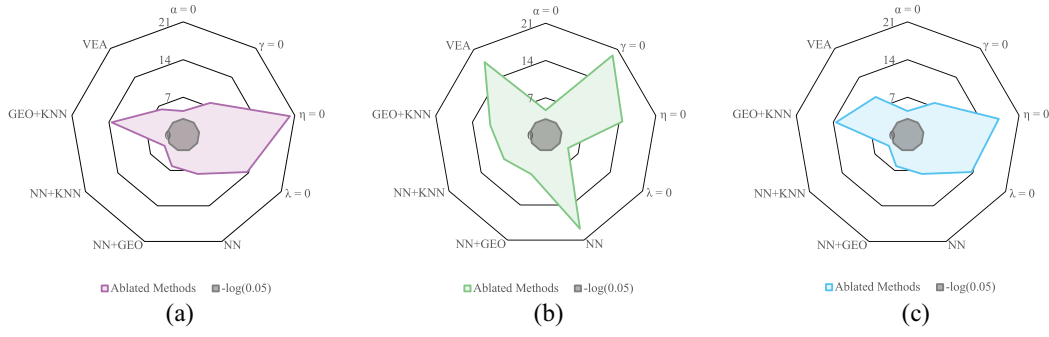


Fig. 9. Significance T -tests on three randomly selected tasks have been performed to verify the statistical soundness of the contributions from different constituting components of the GGA. The gray shaded area denotes the significant threshold $-\log(0.05)$. Among each dimension, the wider the coverage is the more significant the contribution is on that ablated component. (a) $K \rightarrow B$. (b) $N \rightarrow G$. (c) $C \rightarrow W$.

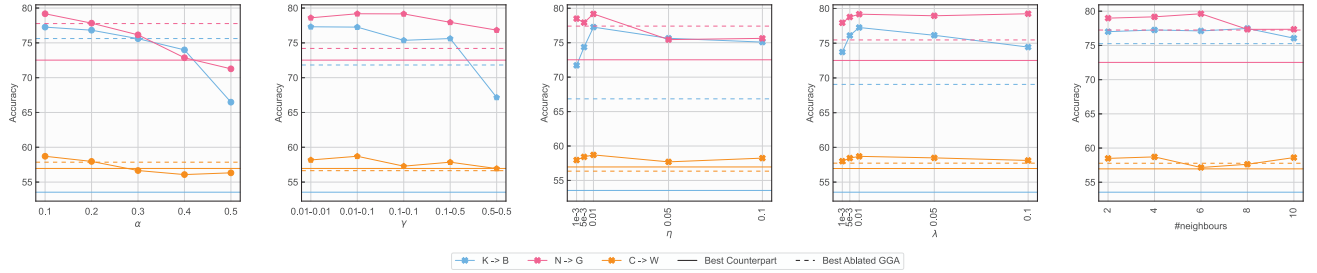


Fig. 10. Parameter sensitivity analysis of the GGA method for hyperparameters α , γ , η , λ , and $\#$ neighbour under their corresponding reasonable range. As shown in the legend, the solid and dashed horizontal lines indicate the best-comparing method and the best-performed GGA ablated counterpart under each task, respectively.

$-\log(0.05)$. Among each dimension, the wider the coverage is, the more significant the contribution is on that ablated component. As we can see from Fig. 9, the colored area has wider coverage than the gray shaded area among all dimensions, which indicates that the contributions made by all constituting components have statistical soundness. Therefore, all proposed components are indispensable for GGA to achieve a fine-grained intrusion knowledge transfer via graph alignment.

I. Parameter Sensitivity

The parameter sensitivity of the GGA method has been illustrated in Fig. 10. The GGA shows relatively stable performance under these hyperparameter ranges without showing severe fluctuations. Besides, the GGA outperforms the best-performed comparing method under nearly all hyperparameter ranges. Additionally, the GGA constantly shows superior performance than its best-ablated counterpart. Therefore, we verify that the GGA method is robust on varied hyperparameter settings.

Besides, during all experiments, only a fixed set of hyperparameters is used to tackle diverse data domains. The GGA can constantly show satisfying performance without the need to perform time-consuming hyperparameter resetting. Therefore, it further demonstrates the robustness of GGA on hyperparameters and its usefulness when tackling diverse intrusion data domains.

J. Computational Efficiency

Finally, to verify the computational efficiency of the GGA method, we measure both the training time per epoch and inference time per instance and make comparison between two

TABLE VII
COMPARISON ON TRAINING TIME PER EPOCH (SECOND)

$S \rightarrow T$	$K \rightarrow B$	$N \rightarrow G$	$C \rightarrow W$	Avg
DDAC	7.54	3.86	1.96	4.45
WCGN	0.18	0.11	0.15	0.15
GGA (Ours)	0.17	0.11	0.13	0.14

TABLE VIII
COMPARISON ON INFERENCE TIME PER INSTANCE
(MICROSECOND = 10^{-6} SECOND)

$S \rightarrow T$	$K \rightarrow B$	$N \rightarrow G$	$C \rightarrow W$	Avg
DDAC	1.43	1.76	0.96	1.38
WCGN	0.20	0.17	0.19	0.19
GGA (Ours)	0.17	0.14	0.17	0.16

best-performed baseline counterparts. The results are presented in Tables VII and VIII. As we can observe, the GGA achieves the best training and inference efficiency. Specifically, the GGA trains 31 times and 6.67% faster than DDAC and WCGN, respectively. Besides, the GGA also achieves the fastest inference speed, which outperforms the best-performed counterpart WCGN by 15.79%. Hence, it verifies the efficiency of the GGA, making it suitable to be used under computationally-constrained IoT scenarios.

VI. CONCLUSION

In this article, we utilize the knowledge-rich NI domain to facilitate accurate intrusion detection for the data-scarce IoT domain. We tackle this HDA problem through a GGA approach. First, a PLE mechanism is employed to exploit the unlabeled target instances, which jointly considers the network prediction, geometric property, and neighborhood

TABLE IX
ACRONYM TABLE

Acronym	Interpretation
IID	IoT Intrusion Detection
NID	Network Intrusion Detection
GGA	Geometric Graph Alignment
DA	Domain Adaptation
NI	Network Intrusion
II	IoT Intrusion
PL(s)	Pseudo Label(s)
HDA	Heterogeneous Domain Adaptation
PLE	Pseudo Label Election
WAM(s)	Weighted Adjacency Matrix(Matrices)
NN	Neural Network

TABLE X
NOTATION TABLE

Notation	Interpretation
$\mathcal{D}_*$	Domain, $* \in \{S, TL, TU\}$
x_{*i}	The i^{th} instance of domain $*$
y_{*i}	The intrusion class of the i^{th} instance of domain $*$
n_*	Number of instances in domain $*$
d_*	Dimension of domain $*$ instances
K	Number of intrusion categories
G_X	Domain graph of domain X
V_X	Vertices in domain graph G_X
E_X	Edges in domain graph G_X
V_X^i	Class i vertices in domain graph G_X
E_X	Feature projector for domain X
$f(x_i)$	Features projected by the feature projector
PL_G^i	The geometric label for the i^{th} unlabelled target instance
$\mathcal{X}_S^{(k)}$	Source domain instances belong to class k
$\mu_{S+TL}^{(k)}$	Centroid of the k^{th} class of source and labelled target domain instances
M_X	The weighted adjacency matrix of domain X
$\mathcal{L}_{SK}$	Shape keeping loss
$D()$	The discriminator
$\mathcal{L}_R$	Rotation avoidance loss
$\mathcal{L}_{CP}$	Centre point matching loss
$q^{(k)}$	The correlation semantic of class k for the source domain
$C()$	The shared classifier
T	Temperature hyperparameter controlling the semantic smoothness
p_i	The correlation semantic of the i^{th} labelled target instance
$\mathcal{L}_V$	Vertex-level alignment loss
α	Hyperparameter in $\mathcal{L}_V$
$\mathcal{L}_{ce}$	Cross entropy loss
$\mathcal{L}_{SUP}$	The supervision loss of source domain
γ	Hyperparameter balancing the weight of $\mathcal{L}_{SK}$
η	Hyperparameter balancing the weight of $\mathcal{L}_R$
λ	Hyperparameter balancing the weight of $\mathcal{L}_{CP}$
$TP^{(k)}$	True positive for intrusion category k

information to boost the PL assignment accuracy. The PLE avoids geometrically diverged confident but wrong PLs and near-boundary ambiguous PLs. Then, both intrusion domains are formulated as graphs, with the GGA performed using four mechanisms, from general graph granularity to vertex-level alignment. Specifically, the graph shape is kept via a confused discriminator that is incapable to distinguish the origin of WAMs. Besides, the rotation avoidance mechanism and the center point matching mechanism avoid graph misalignment

caused by rotation and symmetry, respectively. Additionally, the vertex-level semantic is preserved to facilitate a more fine-grained graph alignment. By forming these mechanisms into a holistic whole, the GGA method can align intrusion graphs in a fine-grained manner, which benefits the intrusion knowledge transfer between domains. Comprehensive experiments demonstrate the state-of-the-art performance of the GGA method. Insight analysis also verifies the usefulness of each constituting component of the GGA method.

APPENDIX

Acronym Table: For better readability, we provide the following acronym table for reference. Note that all acronyms are defined in the main text as well as at the first time they are introduced.

Notation Table: We provide a notation table for better readability.

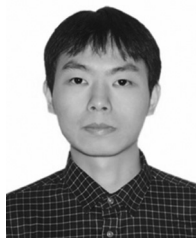
REFERENCES

- [1] M. N. Bhuiyan, M. M. Rahman, M. M. Billah, and D. Saha, "Internet of Things (IoT): A review of its enabling technologies in healthcare applications, standards protocols, security, and market opportunities," *IEEE Internet Things J.*, vol. 8, no. 13, pp. 10474–10498, Jul. 2021.
- [2] J. Wu et al., "Joint semantic transfer network for IoT intrusion detection," *IEEE Internet Things J.*, early access, Nov. 1, 2022, doi: [10.1109/JIOT.2022.3218339](https://doi.org/10.1109/JIOT.2022.3218339).
- [3] C. Dietz et al., "IoT-botnet detection and isolation by access routers," in *Proc. 9th Int. Conf. Netw. Future (NOF)*, 2018, pp. 88–95.
- [4] C. D. McDermott, F. Majdani, and A. V. Petrovski, "Botnet detection in the Internet of Things using deep learning approaches," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2018, pp. 1–8.
- [5] J. Ning et al., "Malware traffic classification using domain adaptation and ladder network for secure industrial Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 17058–17069, Sep. 2022.
- [6] X. Hu, C. Zhu, G. Cheng, R. Li, H. Wu, and J. Gong, "A deep subdomain adaptation network with attention mechanism for malware variant traffic identification at an IoT edge gateway," *IEEE Internet Things J.*, early access, Mar. 21, 2022, doi: [10.1109/JIOT.2022.3160755](https://doi.org/10.1109/JIOT.2022.3160755).
- [7] Z. Wang, Y. Luo, Z. Huang, and M. Baktashmotlagh, "Prototype-matching graph network for heterogeneous domain adaptation," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 2104–2112.
- [8] L. Wang, C. Huang, W. Ma, X. Cao, and S. Vosoughi, "Graph embedding via diffusion-wavelets-based node feature distribution characterization," in *Proc. 30th ACM Int. Conf. Inf. Knowl. Manag.*, 2021, pp. 3478–3482.
- [9] Y.-T. Hsieh, S.-Y. Tao, Y.-H. H. Tsai, Y.-R. Yeh, and Y.-C. F. Wang, "Recognizing heterogeneous cross-domain data via generalized joint distribution adaptation," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, 2016, pp. 1–6.
- [10] J. Li, G. Li, Y. Shi, and Y. Yu, "Cross-domain adaptive clustering for semi-supervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2505–2514.
- [11] Y. Yao, Y. Zhang, X. Li, and Y. Ye, "Heterogeneous domain adaptation via soft transfer network," in *Proc. 27th ACM Int. Conf. Multimedia*, 2019, pp. 1578–1586.
- [12] C. Jun and C. Chi, "Design of complex event-processing IDS in Internet of Things," in *Proc. 6th Int. Conf. Meas. Technol. Mechatronics Autom.*, 2014, pp. 226–229.
- [13] E. Anthei, L. Williams, M. Słowińska, G. Theodorakopoulos, and P. Burnap, "A supervised intrusion detection system for smart home IoT devices," *IEEE Internet Things J.*, vol. 6, no. 5, pp. 9042–9053, Oct. 2019.
- [14] P. Shukla, "ML-IDS: A machine learning approach to detect wormhole attacks in Internet of Things," in *Proc. Intell. Syst. Conf. (IntelliSys)*, 2017, pp. 234–240.
- [15] M. Ge, X. Fu, N. Syed, Z. Baig, G. Teo, and A. Robles-Kelly, "Deep learning-based intrusion detection for IoT networks," in *Proc. IEEE 24th Pac. Rim Int. Symp. Depend. Comput. (PRDC)*, 2019, pp. 256–265.
- [16] Y. Meidan et al., "N-BaIoT—Network-based detection of IoT botnet attacks using deep autoencoders," *IEEE Pervasive Comput.*, vol. 17, no. 3, pp. 12–22, Jul.–Sep. 2018.

- [17] L. Vu, Q. U. Nguyen, D. N. Nguyen, D. T. Hoang, and E. Dutkiewicz, "Deep transfer learning for IoT attack detection," *IEEE Access*, vol. 8, pp. 107335–107344, 2020.
- [18] W.-Y. Chen, T.-M. H. Hsu, Y.-H. H. Tsai, Y.-C. F. Wang, and M.-S. Chen, "Transfer neural trees for heterogeneous domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 399–414.
- [19] Y. Yao, Y. Zhang, X. Li, and Y. Ye, "Discriminative distribution alignment: A unified framework for heterogeneous domain adaptation," *Pattern Recognit.*, vol. 101, May 2020, Art. no. 107165.
- [20] A. Singh et al., "Improving semi-supervised domain adaptation using effective target selection and semantics," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 2709–2718.
- [21] K. Saito, D. Kim, S. Sclaroff, T. Darrell, and K. Saenko, "Semi-supervised domain adaptation via minimax entropy," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 8050–8058.
- [22] T. Kim and C. Kim, "Attract, perturb, and explore: Learning a feature alignment network for semi-supervised domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 591–607.
- [23] M. Pilanci and E. Vural, "Domain adaptation on graphs by learning aligned graph bases," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 2, pp. 587–600, Feb. 2022.
- [24] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by back-propagation," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1180–1189.
- [25] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Comput. Intell. Security Defense Appl.*, 2009, pp. 1–6.
- [26] S. Hettich, "KDD cup 1999 data," Data Set, Dept. Inf. Comput. Sci., Univ. California, Los Angeles, CA, USA, 1999.
- [27] H. M. Harb, A. A. Zaghloul, M. A. Gomaa, and A. S. Desuky, "Selecting optimal subset of features for intrusion detection systems," *Adv. Comput. Sci. Technol.*, vol. 4, no. 2, pp. 179–192, 2011.
- [28] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf. (MilCIS)*, 2015, pp. 1–6.
- [29] O. Alkadi, N. Moustafa, B. Turnbull, and K.-K. R. Choo, "A deep blockchain framework-enabled collaborative intrusion detection for protecting IoT and cloud networks," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9463–9472, Jun. 2021.
- [30] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward generating a new intrusion detection dataset and intrusion traffic characterization," in *Proc. ICISPP*, vol. 1, 2018, pp. 108–116.
- [31] Kurniadi, D. Stiawan, Darmawijoyo, M. Y. B. Idris, A. M. Bamhdi, and R. Budiarto, "CICIDS-2017 dataset feature analysis with information gain for anomaly detection," *IEEE Access*, vol. 8, pp. 132911–132921, 2020.
- [32] N. Koroniotis, N. Moustafa, E. Sitnikova, and B. Turnbull, "Towards the development of realistic botnet dataset in the Internet of Things for network forensic analytics: Bot-IoT dataset," *Future Gener. Comput. Syst.*, vol. 100, pp. 779–796, Nov. 2019.
- [33] T. M. Booi, I. Chiscop, E. Meeuwissen, N. Moustafa, and F. T. H. den Hartog, "ToN-IoT: The role of heterogeneity and the need for standardization of features and attack types in IoT network intrusion data sets," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 485–496, Jan. 2022.
- [34] G. Abdelmoumin, D. B. Rawat, and A. Rahman, "On the performance of machine learning models for anomaly-based intelligent intrusion detection systems for the Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 6, pp. 4280–4290, Mar. 2022.
- [35] H. Qiu, T. Dong, T. Zhang, J. Lu, G. Memmi, and M. Qiu, "Adversarial attacks against network intrusion detection in IoT systems," *IEEE Internet Things J.*, vol. 8, no. 13, pp. 10327–10335, Jul. 2021.
- [36] S. Li, B. Xie, J. Wu, Y. Zhao, C. H. Liu, and Z. Ding, "Simultaneous semantic alignment network for heterogeneous domain adaptation," in *Proc. 28th ACM Int. Conf. Multimedia*, 2020, pp. 3866–3874.
- [37] A. L. Maas, A. Y. Hannun, and A. Y. Ng, "Rectifier nonlinearities improve neural network acoustic models," in *Proc. ICML*, vol. 30, 2013, p. 3.
- [38] J. Li, Z. Zhao, R. Li, and H. Zhang, "AI-based two-stage intrusion detection for software defined IoT networks," *IEEE Internet Things J.*, vol. 6, no. 2, pp. 2093–2102, Apr. 2019.
- [39] M. A. Ferrag, L. Maglaras, S. Moschogiannis, and H. Janicke, "Deep learning for cyber security intrusion detection: Approaches, datasets, and comparative study," *J. Inf. Security Appl.*, vol. 50, Feb. 2020, Art. no. 102419.
- [40] S. Zavrak and M. İskefiyeli, "Anomaly-based intrusion detection from network flow features using variational autoencoder," *IEEE Access*, vol. 8, pp. 108346–108358, 2020.



His research interests include machine learning and cloud computing.



Jiashu Wu (Graduate Student Member, IEEE) received the B.Sc. degree in computer science and financial mathematics and statistics from The University of Sydney, Camperdown, NSW, Australia, in 2018, and the M.IT. degree in artificial intelligence from the University of Melbourne, Parkville, VIC, Australia, in 2020. He is currently pursuing the Ph.D. degree with Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, and the University of Chinese Academy of Sciences, Beijing, China.

Hao Dai (Graduate Student Member, IEEE) received the B.S. and M.Sc. degrees in communication and electronic technology from Wuhan University of Technology, Wuhan, China, in 2015 and 2017, respectively. He is currently pursuing the Ph.D. degree with Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China, and the University of Chinese Academy of Sciences, Beijing, China.

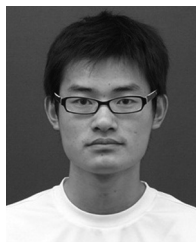
His research interests include mobile edge computing, federated learning and deep reinforcement learning.



Yang Wang (Member, IEEE) received the B.Sc. degree in applied mathematics from the Ocean University of China, Qingdao, China, in 1989, the M.Sc. degree in computer science from Carlton University, Ottawa, ON, Canada, in 2001, and the Ph.D. degree in computer science from the University of Alberta, Edmonton, AB, Canada, in 2008.

He is currently with the Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Beijing, China, as a Professor and as an Adjunct Professor with Xiamen University, Xiamen, China. His research interests include service and cloud computing, programming language implementation, and software engineering.

Dr. Wang was an Alberta Industry Research and Development Associate from 2009 to 2011, and a Canadian Fulbright Scholar from 2014 to 2015.



Kejiang Ye (Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science from Zhejiang University, Hangzhou, China, in 2008 and 2013 respectively.

He was also a joint Ph.D. student with The University of Sydney, Camperdown, NSW, Australia, from 2012 to 2013. After graduation, he worked as a Postdoctoral Researcher with Carnegie Mellon University, Pittsburgh, PA, USA, from 2014 to 2015 and Wayne State University, Detroit, MI, USA, from 2015 to 2016. He is currently an

Associate Professor with the Shenzhen Institute of Advanced Technology, Chinese Academy of Science, Beijing, China. His research interests focus on the performance, energy, and reliability of cloud computing and network systems.



Chengzhong Xu (Fellow, IEEE) received the Ph.D. degree from The University of Hong Kong, Hong Kong, in 1993.

He is currently the Dean of the Faculty of Science and Technology with the University of Macau, Macau, China, and the Director with the Institute of Advanced Computing and Data Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Beijing, China. He has published more than 200 papers in journals and conferences. His research interest

includes parallel and distributed systems, service and cloud computing, and software engineering.

Dr. Xu serves on a number of journal editorial boards, including IEEE TRANSACTIONS ON COMPUTERS, IEEE TRANSACTIONS ON PARALLEL AND DISTRIBUTED SYSTEMS, IEEE TRANSACTIONS ON CLOUD COMPUTING, *Journal of Parallel and Distributed Computing*, and *China Science Information Sciences*.